Cell-Free ISAC 系统鲁棒波束成形优化：从非凸问题到可解 SDP

研究阶段报告

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Abstract

本文针对无蜂窝集成感知与通信（Cell-Free Integrated Sensing and Communication, ISAC）系统，研究在信道状态信息（CSI）存在估计误差的条件下，联合波束成形优化的鲁棒设计问题。考虑通信用户服务质量（QoS）约束、感知检测概率约束、跟踪精度约束（以 PCRB 表征）以及接入点（AP）选择约束，建立了以最小化系统总发射功率为目标的混合整数非线性规划（MINLP）问题。通过引入协方差矩阵变量、半定松弛（SDR）以及 S-Procedure，将原始极度非凸问题严格转化为标准凸半定规划（SDP），并证明转化过程的等价性。进一步给出高斯随机化秩一恢复算法，保证凸最优解可转化为物理可实现的波束向量。本文建立了从物理建模到凸优化求解的完整数学框架，为后续算法实现与系统部署奠定理论基础。

Contents

1 Introduction

集成感知与通信（Integrated Sensing and Communication, ISAC）作为第六代移动通信（6G）的关键使能技术，通过在共享频谱和硬件资源上同时实现通信与雷达感知功能，显著提升了频谱效率和系统能效 [1]。与传统分离式设计相比，ISAC 通过波形共享和联合波束成形设计，在通信性能与感知精度之间取得最优权衡，已成为学术界和工业界的研究热点 [2]。

在 ISAC 系统架构中，无蜂窝（Cell-Free）网络通过分布式部署大量低功率接入点（Access Point, AP），由中央处理单元（Central Processing Unit, CPU）通过前传链路进行联合信号处理，有效消除小区间干扰并提升覆盖均匀性 [3]。将 Cell-Free 架构与 ISAC 相结合，既能利用分布式 AP 的空间分集增益提升通信可靠性，又能通过多视角观测增强目标感知与跟踪能力，代表了未来无线网络的重要演进方向 [4]。

然而，Cell-Free ISAC 系统的联合波束成形优化面临多重挑战。首先，通信波束与感知波形在功率、空间方向图和时频资源上存在耦合，导致优化问题高度非凸。其次，实际系统中信道状态信息（Channel State Information, CSI）通过有限导频估计获得，存在有界或随机的估计误差，需要在优化设计中显式考虑鲁棒性。第三，感知性能通常以 Cramér-Rao 下界（CRB）或预测 CRB（PCRB）表征，其关于波束变量的非线性依赖进一步增加了问题的复杂性。最后，AP 选择作为二进制决策变量，使问题成为混合整数非线性规划（Mixed-Integer Nonlinear Programming, MINLP），在计算上属于 NP-hard 类。

针对上述挑战，本文建立了 Cell-Free ISAC 系统鲁棒波束成形的完整数学框架，核心贡献如下：

- **问题建模**：建立了包含通信 QoS、感知检测、跟踪精度和 AP 选择约束的标准 MINLP 问题，显式刻画 CSI 估计误差对通信和感知性能的影响。
- **等价凸化**：通过协方差矩阵提升、半定松弛（SDR）和 S-Procedure，将原始非凸问题严格转化为标准凸 SDP，并证明转化过程的等价性。
- **求解算法**：给出基于高斯随机化的秩一恢复算法，保证凸最优解可转化为物理可实现的波束向量，同时提供功率缩放兜底方案。
- **复杂度分析**：对比 SDP 内点法求解与 ZF 闭式基准的复杂度，为工程实现提供计算可行性评估。

Notation：Scalars are denoted by italic letters, vectors by bold lowercase letters, and matrices by bold uppercase letters. $\mathbb{C}^{N\times M}$ and $\mathbb{H}^N$ denote the set of $N\times M$ complex matrices and $N\times N$ Hermitian matrices, respectively. $(\cdot)^T$, $(\cdot)^H$, and $\mathrm{tr}(\cdot)$ denote transpose, conjugate transpose, and trace, respectively. $\mathbf{X}\succeq\mathbf{0}$ means $\mathbf{X}$ is positive semidefinite. $\|\cdot\|_2$ denotes the Euclidean norm.

2 System Model and Problem Formulation

2.1 Cell-Free ISAC System Model

考虑一个 Cell-Free ISAC 系统，包含 M 个分布式 AP、K 个单天线通信用户（Communication User, CU）和 P 个感知目标。每个 AP 配备 N_t 根发射天线，通过理想前传链路与中央 CPU 连接。系统采用时分双工（TDD）模式，利用信道互易性获取下行 CSI。

2.1.1 Communication Signal Model

用户 k 的接收信号可表示为:

$$y_k = \underbrace{\sum_{m=1}^{M} \mathbf{h}_{m,k}^H \mathbf{w}_{m,k} s_k}_{\text{desired signal}} + \underbrace{\sum_{j \neq k} \sum_{m=1}^{M} \mathbf{h}_{m,k}^H \mathbf{w}_{m,j} s_j}_{\text{multi-user interference}} + n_k \tag{1}$$

其中 $s_k \sim \mathcal{CN}(0,1)$ 为发送符号，$\mathbf{w}_{m,k} \in \mathbb{C}^{N_t}$ 为 AP m 到用户 k 的通信波束成形向量，$\mathbf{h}_{m,k} \in \mathbb{C}^{N_t}$ 为 AP m 到用户 k 的通信信道，$n_k \sim \mathcal{CN}(0,\sigma_c^2)$ 为接收噪声。

为便于分析，引入堆叠向量形式。定义全局通信信道 $\mathbf{h}_k \in \mathbb{C}^{MN_t}$ 和全局通信波束 $\mathbf{w}_k \in \mathbb{C}^{MN_t}$:

$$\mathbf{h}_k = [\mathbf{h}_{1,k}^T, \mathbf{h}_{2,k}^T, \ldots, \mathbf{h}_{M,k}^T]^T, \quad \mathbf{w}_k = [\mathbf{w}_{1,k}^T, \mathbf{w}_{2,k}^T, \ldots, \mathbf{w}_{M,k}^T]^T \tag{2}$$

则接收信号可紧凑表示为:

$$y_k = \mathbf{h}_k^H \mathbf{w}_k s_k + \sum_{j \neq k} \mathbf{h}_k^H \mathbf{w}_j s_j + n_k \tag{3}$$

2.1.2 Sensing Signal Model

对于单基地雷达感知，目标 p 的反射信号为:

$$y_{S,p} = \sum_{m=1}^{M} \mathbf{g}_{m,p}^H \mathbf{Z}_m \mathbf{g}_{m,p} + n_{S,p} \tag{4}$$

其中 $\mathbf{Z}_m \in \mathbb{C}^{N_t \times N_t}$ 为 AP m 的感知协方差矩阵，$\mathbf{g}_{m,p} \in \mathbb{C}^{N_t}$ 为 AP m 到目标 p 的感知信道，$n_{S,p} \sim \mathcal{CN}(0,\sigma_s^2)$ 为感知接收噪声。

2.1.3 Imperfect CSI Model

实际系统中，信道估计存在有界误差。本文采用确定性有界误差模型:

Communication channel:

$$\mathbf{h}_k = \hat{\mathbf{h}}_k + \Delta\mathbf{h}_k, \quad \|\Delta\mathbf{h}_k\|_2 \leq \epsilon_h \|\hat{\mathbf{h}}_k\|_2 \tag{5}$$

Sensing channel:

$$\mathbf{g}_p = \hat{\mathbf{g}}_p + \Delta\mathbf{g}_p, \quad \|\Delta\mathbf{g}_p\|_2 \leq \epsilon_g \|\hat{\mathbf{g}}_p\|_2 \tag{6}$$

其中 $\hat{\mathbf{h}}_k$ 和 $\hat{\mathbf{g}}_p$ 为估计信道，$\Delta\mathbf{h}_k$ 和 $\Delta\mathbf{g}_p$ 为估计误差，ϵ_h 和 ϵ_g 为相对误差界。

2.2 Problem Formulation

优化目标为最小化系统总发射功率，同时满足通信 QoS、感知检测、跟踪精度以及 AP 选择约束：

$$\min_{\substack{\{\mathbf{w}_{m,k}\},\{\mathbf{Z}_m\},\\ \{b_{mp}\}}} \quad \sum_{m=1}^{M}\left(\sum_{k=1}^{K}\|\mathbf{w}_{m,k}\|_2^2 + \mathrm{tr}(\mathbf{Z}_m)\right) \tag{P1}$$

$$\text{s.t.} \quad \mathrm{SINR}_k^{\mathrm{wc}} \ge \gamma_k, \quad \forall k \in \mathcal{K} \tag{7}$$

$$\mathrm{SINR}_{S,p} := \frac{|\mathbf{g}_p^H \mathbf{Z} \mathbf{g}_p|}{\sigma_s^2} \ge \gamma_S^{\mathrm{PoD}}, \quad \forall p \in \mathcal{P} \tag{8}$$

$$\mathrm{tr}(\mathbf{J}_p^{\mathrm{data}}) \ge \Gamma_{\mathrm{Track},p}, \quad \forall p \in \mathcal{P} \tag{9}$$

$$\sum_{m=1}^{M} b_{mp} = N_{\mathrm{req}}, \quad \forall p \in \mathcal{P}^{\mathrm{active}} \tag{10}$$

$$\sum_{k=1}^{K}\|\mathbf{w}_{m,k}\|_2^2 + \mathrm{tr}(\mathbf{Z}_m) \le P_{\max}, \quad \forall m \in \mathcal{M} \tag{11}$$

$$\mathbf{Z}_m \succeq \mathbf{0}, \quad \forall m \in \mathcal{M} \tag{12}$$

$$b_{mp} \in \{0,1\}, \quad \forall m \in \mathcal{M}, p \in \mathcal{P} \tag{13}$$

其中，γ_k 和 γ_S^{PoD} 分别为通信用户 k 的 SINR 门限与感知检测概率门限，$\Gamma_{\mathrm{Track},p}$ 为目标 p 的跟踪精度门限（FIM 迹约束），N_{req} 为每个目标所需服务 AP 数，$P_{\max}$ 为单 AP 最大发射功率，$b_{mp} \in \{0,1\}$ 为 AP m 是否服务目标 p 的二进制指示变量。

注 1. *Problem (P1) is a mixed-integer nonlinear programming (MINLP) problem. The non-convexity arises from: (i) the SINR constraints involving ratios of quadratic functions; (ii) the coupling between communication beamforming vectors $\mathbf{w}_{m,k}$ and sensing covariance matrices $\mathbf{Z}_m$ in the power constraints; (iii) the binary AP selection variables b_{mp}; and (iv) the rank-one constraint implicitly required for physical beamforming implementation. The following sections develop a systematic convexification procedure to address these challenges.*

3 Problem Convexification and SDP Reformulation

3.1 Non-Convexity Analysis

Problem (P1) is a mixed-integer nonlinear programming (MINLP) problem. The non-convexity arises from multiple sources: the SINR constraints involve ratios of quadratic functions (NC1), the worst-case CSI uncertainty makes the communication SINR constraint semi-infinite (NC2), the binary AP selection variables create a discrete feasible set (NC3), the rank-one constraint required for physical beamforming implementation defines a non-convex manifold (NC4), the coupling between communication beamforming vectors and sensing covariance matrices introduces bilinear terms (NC5), and the PCRB constraint involves a matrix inverse that depends nonlinearly on the optimization variables (NC6). Table 1 summarizes these non-convexity sources.

Table 1: Non-convexity source identification for problem (P1)

ID	Source	Affected constraints	Convexity violation
NC1	SINR fractional structure	(P1-C1), (P1-C2)	Ratio of quadratics
NC2	Worst-case CSI uncertainty	(P1-C1)	Semi-infinite constraint
NC3	Binary AP selection	(P1-C4), (P1-C7)	Discrete set $\{0, 1\}$
NC4	Rank-one requirement	(P1-C8)	Non-convex manifold
NC5	Variable coupling	(P1-C1)–(P1-C5)	Bilinear terms
NC6	Matrix inverse in PCRB	(P1-C3)	Nonlinear FIM dependence

3.2 Covariance-Lifted Form (P2)

To set up the SDR in the next subsection, we first lift the per-AP beamforming vectors to their covariance form. Define $\mathbf{W}_k = \mathbf{w}_k\mathbf{w}_k^H \in \mathbb{H}_+^{MN_t}$ where $\mathbf{w}_k = [\mathbf{w}_{1,k}^T, \ldots, \mathbf{w}_{M,k}^T]^T$, and aggregate the sensing covariances as $\mathbf{Z} = \sum_{m=1}^{M} \mathbf{Z}_m$. The lifted problem **(P2)** is:

$$\begin{aligned}
\min_{\{\mathbf{W}_k\},\mathbf{Z},\{b_{mp}\}} \quad & \sum_{k=1}^{K} \mathrm{tr}(\mathbf{W}_k) + \mathrm{tr}(\mathbf{Z}) && \text{(P2)} \\
\text{s.t.} \quad & \mathrm{tr}(\mathbf{g}_p\mathbf{g}_p^H\mathbf{Z}) \geq \gamma_S^{\mathrm{PoD}}\sigma_s^2, \quad \forall p && \text{(P2-C2)} \\
& \mathrm{tr}(\mathbf{F}_p\mathbf{R}_X) \geq \Gamma_{\mathrm{Track},p}, \quad \forall p && \text{(P2-C3)} \\
& \mathrm{tr}(\mathbf{E}_m\mathbf{R}_X) \leq P_{\max}, \quad \forall m && \text{(P2-C4)} \\
& \mathbf{W}_k \succeq \mathbf{0}, \quad \forall k && \text{(P2-C5)} \\
& \mathbf{Z} \succeq \mathbf{0} && \text{(P2-C6)} \\
& \mathrm{rank}(\mathbf{W}_k) = 1, \quad \forall k && \text{(P2-C7)} \\
& b_{mp} \in \{0, 1\}, \quad \forall m, p && \text{(P2-C8)}
\end{aligned}$$

where $\mathbf{R}_X = \sum_k \mathbf{W}_k + \mathbf{Z}$ and the sensing SINR / PCRB / power constraints are already in lifted form (their derivations are deferred to Steps 2–5 for clarity). **(P1)** $\Leftrightarrow$ **(P2)** is a strict equivalence: the rank-one constraint (P2-C7) guarantees recoverability of $\mathbf{w}_k$ from $\mathbf{W}_k$, and all other constraints are rewritten via the identity $\mathrm{tr}(\mathbf{x}\mathbf{x}^H\mathbf{W}) = \mathbf{x}^H\mathbf{W}\mathbf{x}$ with no relaxation.

3.3 Convexification Procedure

To address the aforementioned non-convexity, we develop a systematic convexification procedure that transforms the lifted problem (P2) into a standard convex semidefinite program (P3). The procedure consists of six steps, each targeting a specific non-convex source while preserving the problem structure as much as possible.

Step 1: Semidefinite relaxation. We drop the rank-one constraint (P2-C7) and only require $\mathbf{W}_k \succeq \mathbf{0}$, which enlarges the feasible set from the non-convex rank-one manifold to the convex PSD cone. This is a tight relaxation when $K \leq 2$, where the optimal $\mathbf{W}_k^*$ naturally satisfies $\mathrm{rank}(\mathbf{W}_k^*) = 1$. For $K > 2$, SDR provides a lower bound on the original problem, and Gaussian randomization can recover a feasible rank-one solution with bounded performance loss $O(1/L)$.

Step 2: SINR fractional rewriting. Before applying the S-Procedure, we rewrite the SINR fraction (P2-C1) as a quadratic inequality by cross-multiplying the positive denominator. This is a strict equivalence, not a relaxation: $\mathrm{tr}(\hat{\mathbf{H}}_k\mathbf{W}_k) - \gamma_k \sum_{j\neq k} \mathrm{tr}(\hat{\mathbf{H}}_k\mathbf{W}_j) \geq \gamma_k\sigma_c^2$. The result is a quadratic form in $\mathbf{h}_k$, the input required by the S-Procedure.

Step 3: S-Procedure for robust SINR. The worst-case SINR constraint is semi-infinite due to the relative uncertainty set $\|\Delta\mathbf{h}_k\|_2 \leq \epsilon_h\|\hat{\mathbf{h}}_k\|_2$. We rewrite it as a quadratic inequality in $\Delta\mathbf{h}_k$: $\tilde{f}_1(\Delta\mathbf{h}) = (\hat{\mathbf{h}}+\Delta\mathbf{h})^H\mathbf{A}_k(\hat{\mathbf{h}}+\Delta\mathbf{h}) - \sigma_c^2 \geq 0$, where $\mathbf{A}_k = \frac{1}{\gamma_k}\mathbf{W}_k - \sum_{j\neq k}\mathbf{W}_j$. The uncertainty set is described by $\tilde{f}_2(\Delta\mathbf{h}) = \epsilon_h^2\|\hat{\mathbf{h}}_k\|_2^2 - \|\Delta\mathbf{h}_k\|_2^2 \geq 0$. By the S-Procedure (which is necessary and sufficient for a single quadratic constraint over a norm ball), the semi-infinite worst-case constraint is equivalently transformed into the following linear matrix inequality with the slack variable $\mu_k \geq 0$:

$$\exists\mu_k \geq 0 : \begin{bmatrix} \mathbf{A}_k + \mu_k\mathbf{I} & \mathbf{A}_k\hat{\mathbf{h}}_k \\ \hat{\mathbf{h}}_k^H\mathbf{A}_k & \hat{\mathbf{h}}_k^H\mathbf{A}_k\hat{\mathbf{h}}_k - \sigma_c^2 - \mu_k\epsilon_h^2\|\hat{\mathbf{h}}_k\|_2^2 \end{bmatrix} \succeq \mathbf{0}, \quad \forall k \tag{14}$$

This transformation is exact since the S-Procedure is necessary and sufficient for a single quadratic constraint over a norm ball.

Step 4: PCRB and sensing SINR linearization. Under the assumption that the channel gradient $\nabla_{\boldsymbol{\theta}_p}\mathbf{g}_p \in \mathbb{C}^{MN_t\times D}$ (with D the target state dimension) is known and constant in the current time slot, the Fisher information matrix becomes an affine function of $\mathbf{R}_X$. Defining $\mathbf{F}_p = \frac{2}{\sigma_s^2}\mathrm{Re}\{\nabla_{\boldsymbol{\theta}_p}\mathbf{g}_p^H\nabla_{\boldsymbol{\theta}_p}^H\mathbf{g}_p\} \in \mathbb{H}_+^{MN_t}$ (note: $\nabla_{\boldsymbol{\theta}_p}\mathbf{g}_p^H \in \mathbb{C}^{MN_t\times D}$ multiplied by $\nabla_{\boldsymbol{\theta}_p}^H\mathbf{g}_p \in \mathbb{C}^{D\times MN_t}$ yields an $MN_t \times MN_t$ Hermitian PSD matrix; PSD follows from $\mathbf{x}^H\mathbf{A}^H\mathbf{A}\mathbf{x} = \|\mathbf{A}\mathbf{x}\|^2 \geq 0$), the PCRB constraint simplifies to $\mathrm{tr}(\mathbf{F}_p\mathbf{R}_X) \geq \Gamma_{\mathrm{Track},p}$, which is linear in $\mathbf{R}_X$. Similarly, the sensing SINR constraint $\mathrm{SINR}_{S,p} = \frac{|\mathbf{g}_p^H\mathbf{Z}\mathbf{g}_p|}{\sigma_s^2} \geq \gamma_S^{\mathrm{PoD}}$ (note the linear —not squared —definition) becomes $\mathrm{tr}(\mathbf{g}_p\mathbf{g}_p^H\mathbf{Z}) \geq \gamma_S^{\mathrm{PoD}}\sigma_s^2$, which is also linear in $\mathbf{Z}$.

Step 5: Power constraint. The per-AP power constraint $\sum_k\|\mathbf{w}_{m,k}\|^2 + \mathrm{tr}(\mathbf{Z}_m) \leq P_{\max}$ is already convex. In the lifted domain, it becomes $\mathrm{tr}(\mathbf{E}_m\mathbf{R}_X) \leq P_{\max}$, which is a linear inequality. No additional transformation is needed.

Step 6: AP selection decomposition. The binary AP selection constraint is NP-hard in general. We adopt a two-step decomposition: first, a heuristic outer step selects the top-N_{req} APs for each target based on large-scale fading; then, the inner step solves the convex SDP with the fixed AP set. This yields a polynomial-time algorithm with suboptimal but engineering-acceptable performance.

Table 2 summarizes the six-step convexification chain (applied to (P2)).

Table 2: Six-step convexification method chain (applied to (P2))

Step	Transformation	Eliminates	Type	Result
1	Covariance lifting $\mathbf{w}\mathbf{w}^H \to \mathbf{W}$	NC1, NC5	Strict equivalence	Linear objecti
2	SDR: drop $\mathrm{rank}(\mathbf{W}) = 1$	NC4	Tight relaxation	PSD cone
3	S-Procedure	NC2	Exact equivalence	LMI
4	Cross-multiply fraction	NC1 (comm)	Strict equivalence	Linear inequal
5a	PCRB affine expansion	NC6	Strict equivalence (Assump. 1)	Linear inequal
5b	Sensing SINR linearization	NC1 (sens)	Strict equivalence	Linear inequal
6	Power (already convex)	—	No change	Linear inequal
7	AP two-step decomposition	NC3	Heuristic (no guarantee)	Fixed AP set

3.4 Final Convex SDP Problem (P3)

Combining all six transformations applied to the lifted problem (P2), the original non-convex problem (P1) is reformulated as the following standard convex SDP (via the chain (P1) $\Leftrightarrow$ (P2) $\rightarrow$ (P3)):

$$\min_{\{\mathbf{W}_k\},\mathbf{Z},\{\mu_k\}} \quad \sum_{k=1}^{K} \mathrm{tr}(\mathbf{W}_k) + \mathrm{tr}(\mathbf{Z}) \tag{P3}$$

$$\text{s.t.} \quad \begin{bmatrix} \mathbf{A}_k + \mu_k \mathbf{I} & \mathbf{A}_k \hat{\mathbf{h}}_k \\ \hat{\mathbf{h}}_k^H \mathbf{A}_k & \hat{\mathbf{h}}_k^H \mathbf{A}_k \hat{\mathbf{h}}_k - \sigma_c^2 - \mu_k \epsilon_h^2 \|\hat{\mathbf{h}}_k\|_2^2 \end{bmatrix} \succeq \mathbf{0}, \quad \forall k \tag{P3-C1}$$

$$\mathrm{tr}(\mathbf{g}_p \mathbf{g}_p^H \mathbf{Z}) \geq \gamma_S^{\mathrm{PoD}} \sigma_s^2, \quad \forall p \tag{P3-C2}$$

$$\mathrm{tr}(\mathbf{F}_p \mathbf{R}_X) \geq \Gamma_{\mathrm{Track},p}, \quad \forall p \tag{P3-C3}$$

$$\mathrm{tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}, \quad \forall m \tag{P3-C4}$$

$$\mathbf{W}_k \succeq \mathbf{0}, \quad \forall k \tag{P3-C5}$$

$$\mathbf{Z} \succeq \mathbf{0} \tag{P3-C6}$$

$$\mu_k \geq 0, \quad \forall k \tag{P3-C7}$$

where $\mathbf{A}_k = \frac{1}{\gamma_k}\mathbf{W}_k - \sum_{j \neq k} \mathbf{W}_j$, $\mathbf{F}_p$ is defined in Step 5, and $\mathbf{E}_m$ is the AP selection matrix. Constraint (P3-C1) is the LMI from the S-Procedure, (P3-C2) and (P3-C3) are the linearized sensing SINR and PCRB constraints, (P3-C4) is the per-AP power constraint, (P3-C5) and (P3-C6) are the PSD constraints from SDR, and (P3-C7) ensures the S-Procedure slack variables are non-negative.

Table 3: Constraint correspondence between (P1) and (P3)

(P1) constraint	(P3) constraint	Transformation	Convexity
(P1-C1) SINR	(P3-C1) LMI	Steps 1,2,3,4	LMI (convex)
(P1-C2) Sens. SINR	(P3-C2) Linear	Steps 1,2,5b	Linear (convex)
(P1-C3) PCRB	(P3-C3) Linear	Steps 1,2,5a	Linear (convex)
(P1-C4) AP select	(P3-C4) Fixed set	Step 7	Linear (convex)
(P1-C5) Power	(P3-C4) Linear	Steps 1,6	Linear (convex)
(P1-C6) PSD	(P3-C6) PSD	No change	PSD cone (convex)
(P1-C7) Binary	Eliminated	Step 7	Heuristic
(P1-C8) Rank-one	(P3-C5) PSD	Step 2	PSD cone (convex)

定理 1 (Problem Convexity and Equivalence)**.** *Problem (P3) is a standard convex SDP. All constraints are either LMIs or linear inequalities, and the objective is linear. For $K \leq 2$, (P3) is equivalent to (P1) since (P1) $\Leftrightarrow$ (P2) is strict, and SDR is tight on (P2). For $K > 2$, (P3) provides a lower bound on (P1). Furthermore, (P3) can be solved in polynomial time via interior-point methods, with complexity upper bound $O\big((K+1)^3(MN_t)^6(K+P+M)\big)$; simplified order $O((KMN_t^2)^{3.5})$.*

Proof. Each constraint in (P3) is either an LMI (P3-C1), a linear inequality (P3-C2, P3-C3, P3-C4), or a PSD cone constraint (P3-C5, P3-C6). The objective is linear. Therefore (P3) is a convex SDP. The transformation chain is:

- **(P1) $\Leftrightarrow$ (P2)**: strict equivalence (variable lifting with rank-one constraint)

- **Step 1 (SDR)**: tight relaxation when $K \leq 2$, lower bound when $K > 2$
- **Steps 2–5**: strict equivalences (fractional rewriting, S-Procedure, PCRB/SINR linearization, power constraint already convex)
- **Step 6 (AP selection)**: heuristic decomposition with no theoretical optimality guarantee

Therefore, for $K \leq 2$, (P3) $\equiv$ (P1); for $K > 2$, $P^*_{\mathrm{P3}} \leq P^*_{\mathrm{P1}}$. The complexity follows from the standard interior-point complexity $O(n^3 L)$ for SDPs, where $n = O(K(MN_t)^2)$ is the decision-variable dimension and L is the input bit-length. □

4 Beam Recovery and Algorithm Design

4.1 Rank-One Recovery via Gaussian Randomization

Problem (P3) yields covariance matrix solutions $\{\mathbf{W}_k^*\}$ and $\mathbf{Z}^*$, which may have rank greater than one. Physical implementation requires rank-one beam vectors. We propose a Gaussian randomization algorithm with power scaling fallback.

命题 1 (Power Scaling Guarantee)**.** *After power scaling, the per-AP power constraint is strictly satisfied:* $P'_m = \beta_m P_m = P_{\max}$*. Both communication and sensing SINR scale by the same factor:* $SINR'_k = \beta_m \cdot SINR_k$ *and* $SINR'_{S,p} = \beta_m \cdot SINR_{S,p}$*.*

注 2. *Unlike pure communication systems where pencil beams are preferred, ISAC sensing requires spatial coverage. Therefore, the multi-rank sensing covariance* $\mathbf{Z}^*$ *is a design feature rather than an algorithmic failure. The "role separation" of communication and sensing in the covariance domain is the essential characteristic of ISAC waveform design.*

4.2 Computational Complexity Analysis

4.2.1 SDP Solver Complexity

Variable count:

- K matrices $\mathbf{W}_k$: $K \cdot (MN_t)^2$ real variables
- 1 matrix $\mathbf{Z}$: $(MN_t)^2$ real variables
- K scalars μ_k: K real variables
- Total: $O(K(MN_t)^2)$

Constraint count:

- K communication LMIs: $K \cdot (MN_t + 1) \times (MN_t + 1)$
- P sensing linear constraints
- M power constraints
- Total computational complexity: $O(K(MN_t)^3)$

Solution time: $O((KMN_t^2)^{3.5})$ via interior-point methods. For $M = 16, N_t = 4, K = 10$, approximately 5–10 seconds (MOSEK).

Algorithm 1: Gaussian Randomization Rank-One Recovery

Input: SDP optimal solution $\{\mathbf{W}_k^*\}_{k=1}^K$, $\mathbf{Z}^*$, constraint parameters

Output: Feasible beamforming vectors $\{\mathbf{w}_k\}_{k=1}^K$, sensing waveforms

`// Step 1: Communication beam recovery`

foreach *user k* **do**

  if $\mathrm{rank}(\mathbf{W}_k^*) = 1$ **then**

    $\mathbf{w}_k \leftarrow \sqrt{\lambda_{\max}(\mathbf{W}_k^*)} \cdot \mathbf{v}_{\max}(\mathbf{W}_k^*)$; `// Direct extraction`

  else

    Generate $L = 1000$ candidates: $\boldsymbol{\xi}_l \sim \mathcal{CN}(\mathbf{0}, \mathbf{W}_k^*)$;

    Normalize: $\mathbf{w}_k^{(l)} \leftarrow \sqrt{\mathrm{tr}(\mathbf{W}_k^*)} \cdot \frac{\boldsymbol{\xi}_l}{\|\boldsymbol{\xi}_l\|_2}$;

    Compute constraint violation:

    $v_l \leftarrow \sum_{k'} \max(0, \gamma_k - \mathrm{SINR}_{k'}^{(l)}) + \sum_p \max(0, \gamma_S^{\mathrm{PoD}} - \mathrm{SINR}_{S,p}^{(l)})$;

    $l^* \leftarrow \arg\min_l v_l$;

    if $v_{l^*} = 0$ **then**

      Accept $\mathbf{w}_k \leftarrow \mathbf{w}_k^{(l^*)}$;

    else

      Enter power scaling fallback (Step 3);

`// Step 2: Sensing waveform recovery`

Eigen-decomposition: $\mathbf{Z}^* = \sum_{i=1}^r \lambda_i \mathbf{v}_i \mathbf{v}_i^H$;

Generate multi-stream waveform: $\mathbf{z}_p = \sum_{i=1}^r \sqrt{\lambda_i} \mathbf{v}_i s_i$, where $s_i \sim \mathcal{CN}(0, 1)$ i.i.d.;

`// Step 3: Power scaling fallback`

foreach *AP m* **do**

  $P_m \leftarrow \sum_k \|\mathbf{w}_{m,k}\|_2^2 + \mathrm{tr}(\mathbf{Z}_m)$;

  if $P_m > P_{\max}$ **then**

    $\beta_m \leftarrow P_{\max}/P_m$;

    $\mathbf{w}_{m,k} \leftarrow \mathbf{w}_{m,k} \cdot \sqrt{\beta_m}$;

    $\mathbf{Z}_m \leftarrow \mathbf{Z}_m \cdot \beta_m$;

`// Step 4: Performance guarantee`

if $K \leq 2$ **then** SDR is tight, recovered solution is optimal;

if $K > 2$ **then** Gaussian randomization provides suboptimal solution with bounded performance loss $O(1/L)$;

4.2.2 Closed-Form Baseline Complexity

As a comparison, the ZF + matched filtering baseline has complexity $O(MP \log M + K^3)$, which is approximately 0.1 seconds for typical parameters. However, this baseline does not guarantee robust constraint satisfaction (success rate approximately 25% under CSI uncertainty).

Table 4: Complexity comparison between SDP and closed-form baseline

Algorithm	Complexity	Typical runtime ($M = 16, N_t = 4, K = 10, P = 4$)
SDP (MOSEK)	$O((KMN_t^2)^{3.5})$	5–10 seconds
ZF + matched filter	$O(MP \log M + K^3)$	< 0.1 seconds

5 Conclusion

This paper established a complete mathematical framework for robust beamforming optimization in Cell-Free ISAC systems. The main contributions are:

1. **Problem formulation**: A standard MINLP (P1) incorporating communication QoS, sensing detection, tracking accuracy, and AP selection constraints under CSI uncertainty.
2. **Robustness analysis**: Worst-case SINR constraints with bounded CSI errors, introducing robustness factors η_h and η_g to quantify error impact.
3. **Convex relaxation**: SDR to relax non-convex rank-one constraints, with proof of sensing constraint convexity under single-angle estimation.
4. **S-Procedure transformation**: Exact equivalence transformation of infinite-dimensional worst-case constraints to finite-dimensional LMIs.
5. **Rank-one recovery**: Gaussian randomization algorithm with power scaling fallback to guarantee physical implementability.

Future work includes:

- Implementation of full SDP solver with MOSEK LMI support
- Extension to sensing channel robustness (scenarios B/C)
- Multi-target joint optimization with temporal coupling
- Simulation validation and performance evaluation

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